

(Autumn 2018)

Ans to the q. no 4

(a)

We know, according to pigeonhole principle,

If N objects are placed in k boxes, then there's at least one box containing N/k objects

Let, minimum number of students needed to ensure that at least eight students will receive the same grade of four grade $A+$, A , $A-$, F is smallest integer N such that $N/4 > 7$. The smallest such integer is $(4 \times 7) + 1 = 29$. Because if we have 28 students - it's possible that seven students received each grade so that no eight students receive same grade. Thus, 29 is the required minimum number of students.

(b)

The word SUCCESS has 7 letters where repeated letters are 3 S and 2 C

∴ Number of different strings that can be made from the word SUCCESS is

$$= \frac{7!}{3! 2!}$$

$$= \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1}$$

$$= 420$$

Thus, 420 different strings are possible from where 1 is SUCCESS so 419 different strings other than SUCCESS.

(c)

For five or six characters long passwords, number of required passwords are (all passwords with alphanumeric character - all passwords

with only alphabets) , since there must be at least 1 digit .

$\therefore$ For 5 characters, number of passwords

$$= 36^5 - 26^5$$

$$= 48584800$$

For 6 characters, number of passwords

$$= 36^6 - 26^6$$

$$= 1867866560$$

$\therefore$ Total number of passwords possible

$$= 48584800 + 1867866560$$

$$= 1916451360$$

(d)

Derangements: A derangement is a permutation of objects that leaves no object in its original position.

We know. number of derangement of a set with n elements is D_n

$$= n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \cdot \frac{1}{n!} \right]$$

In set $\{9, 55, 80, 95\}$, there are 4 elements

$$\therefore D_4 = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)$$

$$= 24 \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right)$$

$$= 9$$

These are :

$$\{95, 9, 55, 80\}$$

$$\{80, 95, 9, 55\}$$

$$\{95, 80, 55, 9\}$$

$$\{80, 9, 95, 55\}$$

$$\{95, 80, 9, 55\}$$

$$\{80, 95, 55, 9\}$$

$$\{55, 80, 95, 9\}$$

$$\{55, 95, 9, 80\}$$

$$\{55, 9, 95, 80\}$$

Ans to the q. no 3

(a)

There are 26 letters in English alphabet.

If none of the letters can be repeated in a three-letter initial of name, 1st letter will have 26 choices, 2nd letter 25 choices and 3rd letter 24 choices.

$$\begin{aligned}\therefore \text{Number of different three-letter initials} \\ &= 26 \times 25 \times 24 \\ &= 15600\end{aligned}$$

(b)

For a bit-string of length six, we have two choices for each bit (0, 1).

$$\begin{aligned}\therefore \text{For length 6, number of possible bit strings} \\ &= 2^6 \\ &= 64\end{aligned}$$

Similarly,

$$\begin{aligned}\text{For length } 5 &= 2^5 \\ &= 32\end{aligned}$$

$$\begin{aligned}\text{For length } 4 &= 2^4 \\ &= 16\end{aligned}$$

$$\begin{aligned}\text{For length } 3 &= 2^3 \\ &= 8\end{aligned}$$

$$\begin{aligned}\text{For length } 2 &= 2^2 \\ &= 4\end{aligned}$$

$$\begin{aligned}\text{For length } 1 &= 2^1 \\ &= 2\end{aligned}$$

$\therefore$ Number of bit strings of length six or less, not counting empty string

$$= 64 + 32 + 16 + 8 + 4 + 2$$

$$= 126$$

(c)

i) Since the woman chooses a ball randomly out of 10 red balls and 10 blue balls, it's possible that the 1st 10 balls she chooses are of the same colour, and no balls from another colour. So, to ensure that she has at least 3 balls of the same ^{both} colour, she has to choose 13 balls

But if she has to choose ^{at least} 3 balls from any one colour, she should choose n balls such that $n/2 > 2$ on $\lceil n/2 \rceil > 3$ (since there are 2 colours)

$\therefore n = 2 \times 2 + 1 = 5$. Required number of balls to choose is 5.

ii) As said in (i), if she selects 10, it's possible that they might all be red. So, to ensure at least 3 blue balls, she must select $(10+3) = 13$ balls.

Ans 1 (e) 10 to 3 (2) 10

A license plate containing three letters followed by two digits (last digit non-zero) will have 26 choices for 3 letters, 10 choices for first digit, 9 choices for 2nd digit. $\therefore$ Number of different license plates

$$= 26 \times 26 \times 26 \times 10 \times 9$$

$$= 1581840$$

(d) $n = 3$

Pigeonhole principle : If N objects are placed in k boxes, there is at least one box containing at least $\lceil N/k \rceil$ objects.

Among 11 ^{integers} ~~digits~~, ~~if~~ at least two must have same digit. Because, we know that there are 10 ^{so, $k=10$} digits, If number of integers

$$N = 11, \quad \lceil N/k \rceil = \lceil 11/10 \rceil = 2. \quad \text{Thus,}$$

through pigeonhole principle, we can prove among 11 integers at least 2 must have same last digit.

Spring - 2019

Ans to the q. no 4

(a)

There are two basic counting principles. They are:

- 1) The sum rule
- 2) The product rule

The sum rule:

If a task can be done either in n_1 ways or in one of n_2 ways, where none of the n_1 ways is same as any of the n_2 ways, then there are $(n_1 + n_2)$ ways of doing the task.

Example: Let either a student of CSE or a student of CEE is chosen as a representative of the ^{science/engineering} faculty of a university. There are 500 students in CSE and 200 students in CEE. Thus, number of students that can

be chosen are $(500 + 200)$ or 700 possible ways to pick the representative.

The product rule: Suppose a procedure can be broken down into a sequence of 2 tasks. If there are n_1 ways of doing the 1st task and for each of these ways, there are n_2 ways of doing the 2nd task, then there are $n_1 \times n_2$ ways of doing the tasks of procedure.

Example: If the chairs of an auditorium can be labelled with a letter and a positive integer not exceeding 100, since there are 26 letters and 100 integers, number of possible ways to label the chairs is $26 \times 100 = 2600$.

(b)

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(c)

The Inclusion-Exclusion principle :

When two tasks can be done at the same time, we cannot use sum rule to find out number of ways to do the tasks. It will lead to overcount of a task, since both tasks are counted separately including the common parts. Then we have to solve this problem by counting the number of ~~tasks~~ ways to do each task and subtract number of ways to do both tasks.

Example; How many ways a bit string of length 8 can be formed that either start with 1 or end with two bits 00

⇒ Number of bit strings of length 8 that

i) begin with 1 = 2^7 (1 bit fixed)
= 128

ii) end with 00 = 2^6 (2 bits fixed)
= 64

iii) start with 1 and
end with 00 = 2^5 (3 bits fixed)
= 32

∴ Required number of bit strings = $128 + 64 - 32$
= 160

Spring 2022

Ans to the q. no 3

(a)

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(b)

If the computer club has 50 members,
number of ways to elect a general secretary,

assistant general secretary and an executive
(no person for more than one position)

$$= 50 \times 49 \times 48$$

$$= 117600$$

(b- or)

Autumn -22 3(c) - or

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(d)

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Spring 2018

Ans to the q. no 4

(a)

For selecting 3 faculty members from the
mathematics department where there are
9 faculty members. number of ways

$$= {}^9C_3$$

$$= 84$$

For selecting 4 faculty members from computer science department where there are 11 members, number of ways are

$$= {}^{11}C_4$$

$$= 330$$

$\therefore$ Number of ways to form the committee

$$= 84 \times 330$$

$$= 27720$$

(b)

Spring - 19 4 (c)

Bit strings of length 10 that

i) begins with three zeros = 2^7 (3 bits fixed)
 $= 128$

ii) ends with two zeros = 2^8 (2 bits fixed)
 $= 256$

iii) begins with three zeros and ends with two zeros = 2^5 (five bits fixed)
 $= 32$

$\therefore$ Number of bit strings of length 10 that either begins with three 0s or ends with two 0s

$$= 128 + 256 - 32$$

$$= 352$$

(c)

The number of ways to choose r elements from n elements where repetition is allowed

$$\text{is } C(n+r-1, r) = C(n+r-1, n-r)$$

$\therefore$ The number of ways to choose 7 ice creams from 3 different types of icecream is

$$= C(3+7-1, 7)$$

$$= C(9, 7)$$

$$= C(9, 2)$$

$$= \frac{9 \times 8}{1 \times 2}$$

$$= 36$$

$\therefore$ There are 36 different ways to select icecreams.

(d)

Generalized pigeonhole principle: If N

objects are placed in k boxes, there are

at least one box containing at least

$\lceil N/k \rceil$ objects.

Proof :

Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then, total number of objects is at most $k(\lceil N/k \rceil - 1)$ which is less than $k((N/k + 1) - 1) = N$ where the inequality $\lceil N/k \rceil < (N/k + 1)$ has been used. This is a contradiction because there are a total of N objects.

Now,

number of integers given = 5

The number of positive possible integers ^(remainders) when a number is divided by 4 = 4

{0, 1, 2, 3}

$\therefore$ Through pigeonhole principle, $\lceil 5/4 \rceil = 2$ that is at least 2 of the numbers have the same number as remainder.

— x —